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QoS–aware Mesh-based Multicast Routing Protocols in Edge Ad Hoc Networks: Concepts and Challenges

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Multicast communication plays a pivotal role in Edge based Mobile Ad hoc Networks (MANETs). MANETs can provide low-cost self-configuring devices for multimedia data communication that can be used in military battlefield, disaster management, connected living, and public safety networks. A Multicast communication should increase the network performance by decreasing the bandwidth consumption, battery power, and routing overhead. In recent years, a number of multicast routing protocols (MRPs) have been proposed to resolve above listed challenges. Some of them are used for dynamic establishment of reliable route for multimedia data communication. This article provides a detailed survey of the merits and demerits of the recently developed techniques. An ample study of various Quality of Service (QoS) techniques and enhancement is also presented. Later, mesh topology-based MRPs are classified according to enhancement in routing mechanism and QoS modification. This article covers the most recent, robust, and reliable QoS-aware mesh based MRPs, classified on the basis of their operational features, and pros and cons. Finally, a comparative study has been presented on the basis of their performance parameters on the proposed protocols.

CCS Concepts: • **Networks** → **Mobile ad hoc networks**; **Routing protocols**; *Cross-layer protocols*; • **General and reference** → *Surveys and overviews;*

Additional Key Words and Phrases: MANETs, multicast, routing protocol, mesh based, quality of service, survey

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1 INTRODUCTION

Mobile Ad hoc Networks (MANETs) came into existence with an aim to handle the undesirable situations that may occur due to disasters such as earthquakes, floods, and fires or due to human activities such as terrorist attacks, military operations, and so on. MANET is a kind of wireless communication network that does not have any stable infrastructure and central administrative control. In past few years, MANETs have been deployed for other purposes also such as audio/video conferencing, emergency rescue operations, traffic control, connected living, and online lectures [6]. These networks possess some excellent features such as fast deployment, flexibility, robustness, mobility support, and highly dynamic network topology (fading, shadowing, network partition) [2]. In MANET, node can communicate with relay (intermediate) nodes if communicating host nodes are not in it's range (multi-hop routing).

MANET is using edge device for deploying their nodes for communicating and analyzing the data in real-time scenarios. Edge devices are helping to reduce the data communication to the end-host and do the minimal required processing at the localhost. **Artificial Intelligence (AI)** is also used to predict the abnormality in the data to secure the transmission of data. Applications of **machine learning (ML)** and reinforcement learning technologies are increasing day by day to make the user's life easier. Authors are using these approaches for optimization [89], stable route finding, position updation [88], and link failure prediction [90]. Pattern recognition method is useful for analyzing the traffic pattern and making the system learn the model. In the future, if the system detect the similar traffic pattern, then it can warn us. It could be helpful predicting malicious attacks.

MANET is a group of wireless mobile nodes that may act as host as well as router and are able to move arbitrarily. MANET is a self-organized network that can be deployed anywhere, at any time, to support particular conditions. In contrast to MANETs, infrastructure-dependent wireless networks are more reliable and provide **Quality of Services (QoS)** assurance. The unreliability in MANETs occur due to limited battery power, limited bandwidth (channel capacity), heterogeneity, high routing overhead, and unpredictable node mobility. Bandwidth, Delay, Signal Strength, and other metrics are used for QoS assurance in multicast group communication for both data and real-time traffic.

In recent years, multicasting has been greatly appreciated in any type of group communication like audio/video conferencing, video lectures, and connected living. In connected living scenario, multicast networks will be a big advantage for reliable communication and easily deployable on edge devices. MRP communicates datagram to a group of destinations recognized by single multicast address at single transmission time. Multicast transmission helps to improve node energy, congestion on channel capacity, time and resource utilization as compared to **Unicast Routing Protocol (URP)**, in case of transmission of datagram to a group of destinations. Multicasting in MANETs is more complex than wired networks, in terms of node energy and bandwidth. High mobility, low channel capacity, and battery issues attracted attention of many researchers towards MRPs to build robust, reliable, and scalable networks. In recent past, numerous surveys have been published on MRP in MANETs. In Reference [52], the authors have placed routing protocols in two broad categories: MRP based on application dependence and application in-dependence. In

Reference [2], authors classified the existing MRPs into three sub categories according to their layer of nature/work namely, the MAC layer, the Network layer, and the Application layer. In Reference [8], authors classified MRPs based on routing protocol mechanisms and focused on reliable and QoS-aware MRP.

Multicast routing protocols have been improved by the researchers consistently on the basis of various evaluation metrics such as quick route recovery, reliability, improved QoS (less energy consumption, reduce channel capacity utilization), less congestion (interference), improved **Packet Delivery Ratio (PDR)** and end-to-end delay, network lifetime, and, last but not the least, security. Group communication faces multiple issues and challenges such as power management, capital management, synchronization, and routing management [8, 32]. Real-time applications require reliable and stable communication among multicast group members. None of the existing works have given the clear picture of QoS–aware mesh based MRPs and their proposed modifications, as per our knowledge. To protect the secure key in mobile devices, He et al. [86] proposed a novel two-party distributed signing protocol for the identity-based signature scheme in IEEE 1363 standard, which allows two devices to generate a valid signature without revealing the entire private key of the user. Later, Feng et al. [87] proposed a new share conversion protocol and constructed the first multi-party distributed signing protocol for the identity-based signature scheme in IEEE 1363 standard.

In this article, we have explored mesh topology-based MRPs and classified them on the basis of modifications in the routing mechanism and QoS metrics adaptation. We have explained major issues and challenges in designing of MRPs such as energy efficiency, reliability, security, and QoS-aware multicasting [62]. We have presented a taxonomy of mesh-based MRPs on the basis of their techniques, features, modification components, and improvement parameters. We have also provided a comparative summary between proposed enhancements on **On Demand Multicast Routing Protocol (ODMRP)** protocol in Table 1. We have presented our proposed routing protocols to resolve the major issues and challenges by creating a reliable routes.

The rest of the article is structured as follows: Current state of MRPs is described in Section 2. We further describe the proposed modifications in mesh-based protocol in Section 3. Section 4 elaborates the multiple mechanisms and requirement of estimating link stability. Section 5 discusses the design issues and challenges in multicast routing protocol. We have added a section for multiple proposed works and their results for overcoming the described challenges presented in Section 6. The future directions and conclusion are given in Section 7 and Section 8, respectively.

2 TAXONOMY OF MULTICAST ROUTING PROTOCOL

Wired network is still the simplest way to use Internet. Nowadays, mobile and portable devices demand internet connectivity at home, at work, and on walk. Wireless networks are categorized as infrastructure and infrastructure-less networks. Cellular network is an example of infrastructure network, with high set-up cost and time. Ad hoc network is an example of infrastructure less network with cost-effectiveness and less set-up time.

Ad hoc means "for the purpose," self-organizing network architecture. There is no requirement of base station. Ad hoc networks are further classified as MANETs, **Wireless Sensor Networks (WSNs)**, **Vehicular Ad hoc Networks (VANETs)** [76, 79, 85], **Wireless Mesh Networks (WMNs)** [78]. Here, we focus on improvement in routing protocols for MANETs [75]. MANETs routing protocols can be classified on the basis of mechanism as reactive (on demand routes are created), proactive (routing table stored the pre-determined routes and update periodically), and hybrid (some nodes have predefined and some have on-demand). In routing mechanism, number of destinations can be classified as Unicast (only one destination supported), Broadcast (to

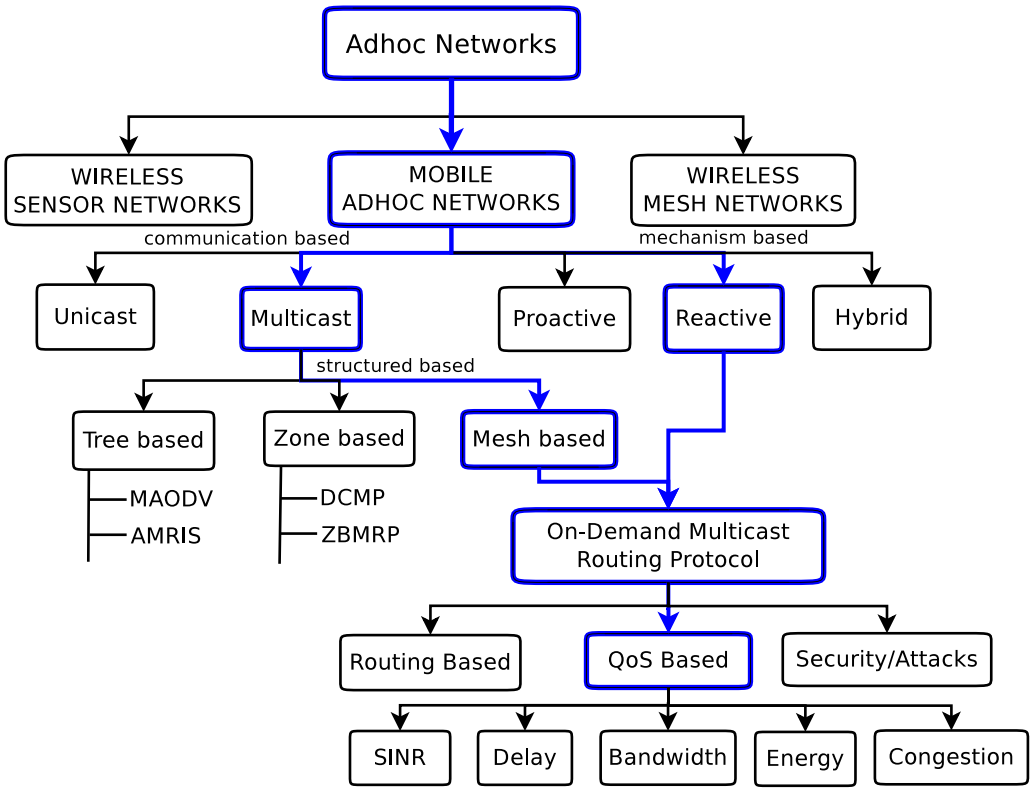

Fig. 1. Taxonomy of routing mechanism in ad hoc networks.

all), or Multicast (for specific group of destinations). Taxonomy of MANETs routing protocols is presented in Figure 1.

Nowadays, **Artificial intelligence (AI)** applications are enhancing the requirement into everyday life such as network traffic data analysis, self-driving cars, entertainment, recommendation systems, health-care, corridor segmentation [91], and agriculture [92]. These connected wireless devices are generating huge volumes of data that should be captured and processed in real time on either edge or server. AI can help in multiple aspects, the major benefits are scalable, adaptive, training from experiences, and ability to find the pattern without requiring an expert in statistics. In ad hoc networks, researchers are applying supervised, unsupervised, and reinforcement learning approach to get the generalize solution for a specific problem that is a bit tricky to solve with mathematical modeling.

Marina et al. [35] explored diverse unicast routing protocol that support QoS. These routing protocols provide a stable communication in unicast routing mechanism. Nowdays, a need to explore Multicast MRPs [8, 16, 17, 36] can help us to avoid the limitations of unicast routing protocols. MRPs can be classified on the basis of their routing structure as (1) tree topology-based, (2) zone topology-based, (3) mesh topology-based, (4) hybrid. Tree topology-based MRP is far efficient in routing for ad hoc network and provides better throughput in terms of packet reception as compared to other protocols, but there is excessive reconfiguration overhead in case of re-routing. In tree-based MRP, there are multiple existing protocols such as **Multicast Ad hoc On demand Distance Vector routing (MAODV)** [25], **Ad hoc Multicast Routing protocol utilizing**

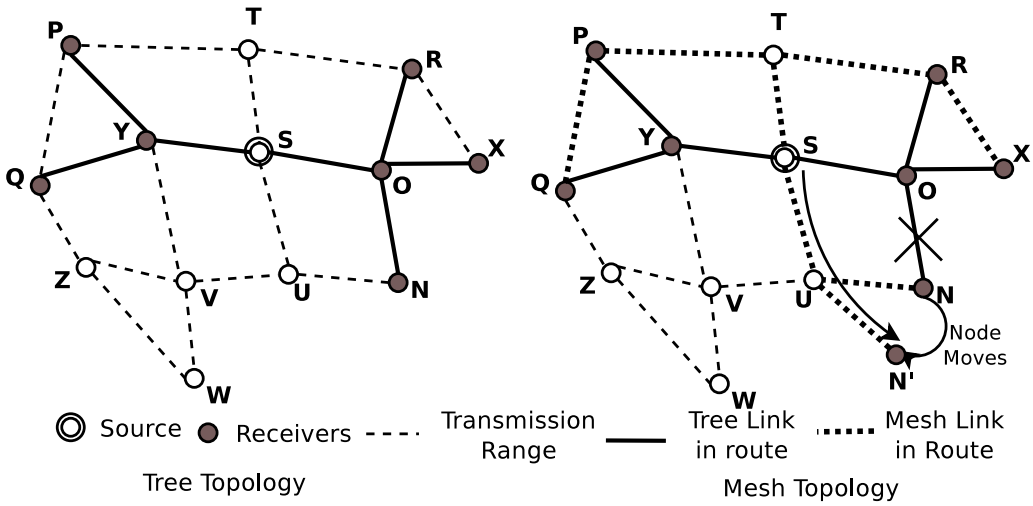

Fig. 2. Difference between tree and mesh topology.

increasing Id numberS (AMRIS) [22], **Ad hoc Multicast Routing protocol (AMRoute)** [23], and so on.

For limited redesigning and reconstructing caused by duplicacy of packets, mesh-based MRP works better than other topologies [8]. Number of paths exist between a pair of source and destinations in mesh-based MRP. There are many protocols such as **On-Demand Multicast Routing Protocol (ODMRP)** [26], **Enhanced On-Demand Multicast Routing Protocol (EODMRP)** [27], **Forwarding Group Multicast Protocol (FGMP)** [38], and **Team Oriented Multicast (TOM)** Protocol [24] that create mesh structure after route construction.

Zone-based MRP forms a cluster of source, receiver, and intermediate nodes on routes and selects zone leader on the basis of first announcement for better and robust decision. There are many protocols that provide zone-based structure for transmission such as **Dynamic Core-based Multicast routing Protocol (DCMP)** [10], **Cluster-based Stable multicast Routing Protocol (CB-SRP)** [12].

To better understand the topological difference between tree and mesh topologies, we have considered a scenario in which one sender and five receivers are there, as shown in Figure 2. The major difference in between tree and mesh topologies is the number of alternative paths provided between two nodes. The former provides only one path, while the latter provides multiple paths for single destination. Due to single path option, tree topologies are not suitable for applications in which link failures and node movement exist. For example, source S is transmitting data to multiple receivers. If link between O and N fails or node N moves to N', then tree topology cannot continue transmission. But mesh topology provides another route to transmit the data due to multiple or alternate paths available between source and destinations. In this survey, we have focused on mesh-based MRPs and explained multiple enhancements in ODMRP protocol on the basis of QoS and routing modification.

3 MESH-BASED MULTICAST ROUTING PROTOCOLS

Mesh topology is reliable and robust for communicating data to the destination in case of node mobility or link failure. It does not require reconfiguration of network because there already exist redundant (multiple) paths for every destination [93].

All forwarding group members, multicast group members, sources and links between them, forms a mesh topology. The characteristic feature of mesh topology is that the node does not care about upstream node, from which source the packet has been arrived, and it rebroadcasts the non-duplicate packets. If one node lies in the communication range of other node, then both nodes share a mesh link [82]. So, the mesh structure has more connected links than tree and increases the robustness of multicast group, which is convenient in generous and frequent link breaks for ad hoc networks [19, 21]. ODMRP protocol robustness depends upon number of sources and their mobility speed. At the low mobility speed and large number of sources, ODMRP established duplicate routes, some of which may be useless; while at high mobility and less number of senders, it reduces the duplicate routes [20].

Forwarding Group Multicast Protocol (FGMP) *for multi-hop, Mobile Wireless Networks [38].* FGMP provides reliability by transmitting data via Forwarding Group and maintains a multicast mesh. In this protocol, both the source and receivers publicize their existence through respective broadcasting packets known as Source Join broadcast (FGMP-SA) Approach and Receiver Join broadcast (FGMP-RA) Approach. When a targeted node retrieves a Join Request from another node, the node updates its own Join table and broadcasts it to other members of group to update their respective table. FGMP reduces overhead by limiting flooding within Forwarding Group. FGMP-SA provides better throughput as compared to FGMP-RA in case of less number of sources than destinations in a network. FGMP protocol is not scalable and it does not support high mobility, because it gives better results in small network.

Core-Assisted Mesh Protocol (CAMP) *[14].* It has been designed to support multicast routing protocol in mobile Ad hoc network using a shared mesh structure. To limit the control traffic, CAMP uses core node for creating a mesh. To avoid duplicate packet or looping in the mesh, every sensor node maintains a cache to keep record of recently forwarded packets. The algorithm ensures that all the nodes from reverse shortest path are included in the mesh. Like other core-based protocols, it does not require whole traffic flow from core nodes. CAMP is based on salient assumption about route information available (proactive) and existence of beaconing protocol. So, it has high overhead because of proactive protocol [15].

3.1 On-Demand Multicast Routing Protocol

Sung et al. [26] have proposed **On-Demand Multicast Routing Protocol (ODMRP)**, a reactive mesh-based Ad hoc multicast routing protocol that gives reliable routes. A complete flow of ODMRP has been illustrated in the Figure 3. One sender and two receivers (nodes with double rings) has been identified in the topology for the communication of data and control packets. To transfer data from the minimal path between source and destination or multicast group address, forwarding group node will be used. In case of route disruption due to node mobility, other shortest path will be used to transmit the data. Mesh topology is more prone to communication disruption due to available multiple paths in mesh structure. A quick example, path $A \rightarrow G \rightarrow K \rightarrow L$ is established as soon as $A \rightarrow G \rightarrow J \rightarrow L$ is disconnected due to node J walk away from the node's communication range.

ODMRP is widely used protocol for group communication in multicast routing protocol due to major advantages of high packet delivery ratio with some limitations such as higher control overhead and redundancy of packet. So, scalability issues occur in ODMRP. Many modifications have been applied on ODMRP [26] to improve the routing overhead. We can classify mesh-based multicast routing protocol based on (1) modified routing mechanism and (2) on adding QoS parameter for improvement in ODMRP protocol. A few number of publications also extended in terms of security analysis and enhancement [11, 34].

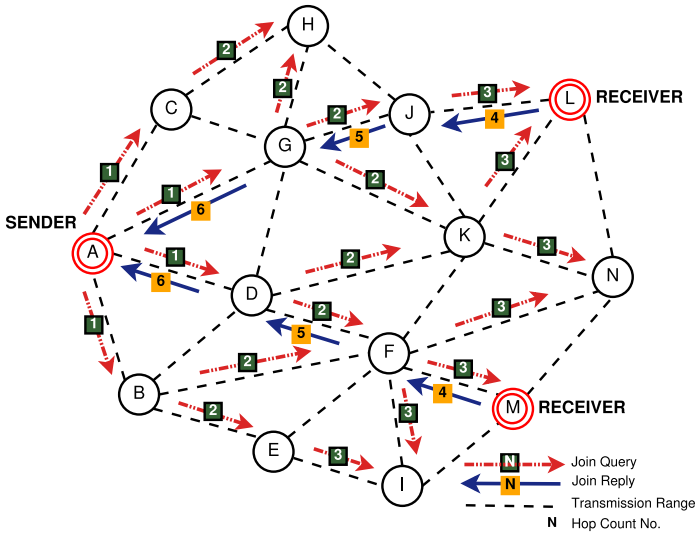

Fig. 3. ODMRP workflow.

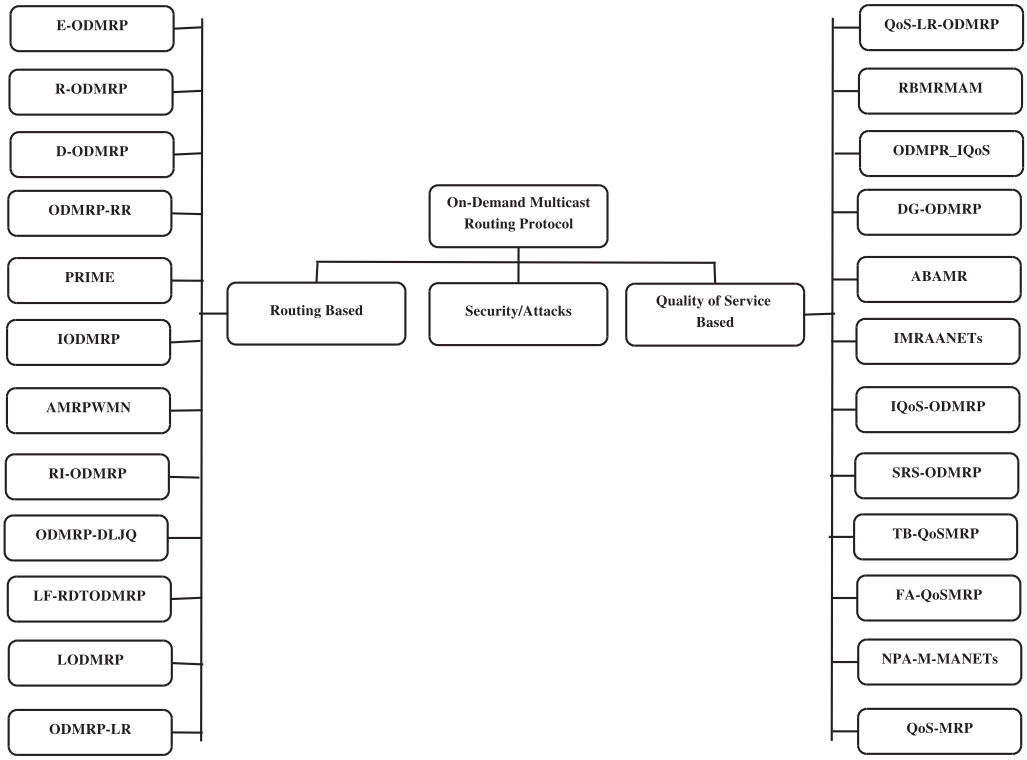

Fig. 4. Enhancement in ODMRP on routing and QoS-based.

3.2 Routing-based Extensions

Routing mechanism in MRP is modified to make it more reliable and robust in terms of packet delivery ratio, end-to-end delay, control overhead, and traffic load. Enhancements in the base ODMRP protocol are based on following routing modification approach:

(1) Local Route Repair: This mechanism is used to avoid global broadcast of messages in case of route or link failure. Only broken link can demand for route and repair it by local recovery mechanism.
(2) Receiver Joining: In this mechanism, if any new incoming destination wants to join current route, it can request for a route from nearby forwarding node, multicast group, or source by broadcasting request packet.
(3) Dynamic Timer Adaption: Motion adaptive refresh interval is utilizing link breakage report to source by receiver. Receiver can make adaptive interval according to their average link lifetime in route to make reliability [70].
(4) Periodic Hello: Periodic Hello packet is broadcast between nodes to extract neighbors' information or link quality.
(5) Route Discovery Suppression: It is used for limiting the number of simultaneous route discoveries as another discovery in process. **Route Discovery Suppression (RDS)** helps us to reduce load on network.
(6) Conserving FG joining: In this mechanism, omit the joining of excessive number of Forwarding Group nodes in route to reduce overhead.

These multiple mechanisms have been used to improve mesh-based ODMRP protocol. In Figure 4, we have listed routing modification protocols over ODMRP protocol.

3.2.1 ODMRP-MPR: On-Demand Multicast Routing Protocol with MultiPoint Relay in MANETs [21]. ODMRP-MPR inducts a multi-point forwarding techniques to mitigate the control overhead, obtain the high stability, and effectively solve the non bidirectional link problem of wireless communication. In the network, each **node (N)** selects few neighbors on the basis of their distance from N and decides two hops as its **multipoint relay (MPR)**, only those neighbors will re-transmit the flooding packet broadcast by N.

ODMRP-MPR reduces flooding overhead generated by Join Query, re-transmission of Join Reply, avoids uni-directional link in forwarding path. But it increases additional overhead by sending periodic Hello messages [83]. NS2 simulator has been used for simulation and compared the control overhead and PDR with varying number of senders and multicast group size with ODMRP protocol.

3.2.2 RODMRP: Resilient On Demand Multicast Routing Protocol [43]. The authors have offered more reliable forwarding path in case of node or route failure in mobility. The redundant packet forwarding improves PDR, while eliminating the possibility of flooding in networks. They create **Non-Forwarding node (NFG)**, which is not a member of Forwarding Group. It is further characterized into active non-forwarding node and passive non-forwarding node. Active non-forwarding nodes forward the data packet in network to improve the degradation in performance caused by node failure. It finds improvement in packet delivery ratio as compared with ODMRP on NS2 simulator.

3.2.3 EODMRP: Enhanced ODMRP with Motion Adaptive Refresh [27]. Dynamic adaptability to refresh rate to the environment and receiver joining is major contribution of this work. Receiver initiates a join query (Receiver Join Query) in the network to join a multicast group. If a route exists to Multicast Receiver or Forwarding Group member, then they should reply with Receiver

Join Reply. Receiver increases the **TTL (Time to live)** value and repeats process until the upper limit of TTL reaches. In worst-case scenario, if no route is found, then the receiver floods a refresh request packet.

This approach has compares the variation of PDR and control overhead over number of receivers on Qualnet simulator. Simulations show that E-ODMRP achieves higher PDR. Protocol has some limitations, because it uses dummy packets and transmit to a sub-tree to prevent recovery explosion, which may result in extra overhead. It also increases extra routing overhead by sending **Receiver Join Query (RJQ)** packet by receiver to join current route and needed additional processing power. Attacker can also waste their resources by sending numerous RJQ requests.

3.2.4 AMRPWMN: Adaptive Multicast Routing Protocol for Wireless Mobile Ad hoc Networks [51]. It improves control packet overhead by broadcasting Join-Query packets according to the current PDR. Due to excessive network overhead and collision, the protocol uses PDR to evaluate Join Query transmission. And, sender in the network broadcasts join query packet according to probability variable, which is calculated by PDR. If PDR is high, then AMRPWMN can transmit more Join Query, else there would be much collision in network. Simulation has been carried out on Glomosim 2.03 simulator and effect of variation of number of senders on different packet delivery ratio has been analyzed.

3.2.5 LF-RDTODMRP: A Robust and Efficient On-demand Multicast Routing Protocol for Ad hoc Network [20]. This protocol cuts down the unnecessary redundant routes and their data transmission. They limit some nodes to flood Join-Request packets and forbid it to be a forwarding node. It adds a data structure (**RDT (Raw Data Table)**) for each forwarding node to make entry of multicast group, source address, and time of entry. RDT table is used for reducing data transmission by using older route.

LF-RDTODMRP limits the flood requests of JQ packet. It adds a Load_Table for storing number of times FG_Flag has been set by multicast group. The protocol sets a threshold value for FG_Flag. If the sum is greater than threshold, then it drops the JQ packets. It sets the threshold value adaptive to the network for better output. This protocol has limitation of Hard to Selection of threshold for number of times FG_Flag has been set. Simulations have been carried out on Glomosim Simulator and obtained results of RDTODMRP, LF-RDTODMRP, and ODMRP are compared based on PDR, Delay, and Overhead with varying traffic load and number of senders.

3.2.6 ODMRP-LR: ODMRP with Link Failure Detection and Local Recovery Mechanism [18]. It addresses the problem of detecting link breakages and local recovery procedure in ODMRP and tries to improve the disadvantage of E-ODMRP by reducing routing overhead. Two methods have been used for detecting link failure: First, by utilizing the knowledge of time intervals between data packets that are to be received. Second, by using Hello packets or data packets in predefined interval. In Glomosim simulator, results have been analyzed for PDR and control overhead on varying TTL value with mobility or non-mobility.

3.2.7 RBMRMAM: Relay-Based Multicast Routing in Multirate-aware MANETs [68]. It minimizes the total transmission time by extracting higher transmission rate of relay node. It proposes **Heuristic Relay Node Selection Algorithm (HRNSA)** for choosing neighbor node that can use higher rate to cover more number of downstream node. It includes three modules: Information collection algorithm, Relay node selection algorithm, and Relay notice algorithm. Throughput and delay are estimated with increasing number of nodes and speed of nodes and compared with that of ODMRP protocol on NS2 Simulator.

3.2.8 ODMRP-DLJQ: Improving Performance of On-Demand Multicast Routing by Deleting Lost Join Query Packet [54]. This work improves the performance by restricting the domain of Join Query packet, which has been lost. It can be achieved by augmenting (increasing) the join query packet with minimum extra information that denotes the number of visited nodes from previous forwarding group nodes. If the current JQ visited many nodes and does not get any previous FG node, then discard it. It reduces overhead with increasing number of forwarding group and hop count over Glomosim Simulator.

3.2.9 ODMRP-RR: Multicast Routing Protocol for Reduction of Relay Node in MANETs [47]. It reduces network overload by reducing number of relay nodes to enhance the performance of ODMRP. It tries to reduce the number of relay nodes that are used for constructing the route and uses Round Robin scheduling for route construction over multiple sources. All sources are not sending join query packets simultaneously. It uses round robin mechanism to differentiate between different sources and allots them distinct time slots. It computes average number of FG nodes for different number of source nodes on NS2 Simulator and Physical testbed.

3.2.10 PRIME: Interest-Driven Approach to Integrated Uni-Multicast Routing [66]. PRIME establishes meshes that are activated and deactivated by the presence or absence of interest in destinations and groups. PRIME establishes enclaves for flows of interest on-demand and proactively sends signals to update routing information within enclaves. Region of network with interest in the destination of flows receives timely updates as compared to other networks.

Meshes are activated using **Mesh-activation Request (MR)**, which makes receivers change their state from inactive to active state. The destination must start advertising its presence periodically using **Mesh Announcements (MAs)**. An enclave of multicast flow is connected components that contain those node dissemination of information for flow. It analyzes group delivery ratio and delay with increasing number of group in Multicast and Unicast traffic for AODV, ODMRP, and PUMA on Qualnet 3.9 Simulator.

3.2.11 RI-ODMRP: Receiver Initiated Mesh-based Multicasting for MANET using ACO [55]. RI-ODMRP approach has been designed to find optimum paths between two communicating nodes. It initializes request by the node that wants to join the multicast group. An Ant Colony-based mechanism is used for multicast routing protocol. Initialize/requesting node is named as core and defines role of node by binary number 11, 01, 10, 00, where first byte is for forwarding group node and second one is for multicast group node. The process of route set up is performed in three steps: (1) Multicast Group Announcement, (2) Multicast Group Joining, and (3) Join Reply. It evaluates average robustness and packet delivery ratio with mobility speed and compared with ODMRP on NS2 Simulator.

3.2.12 D-ODMRP: A Destination-driven ODMRP for MANETs [42]. To improve the multicast forwarding efficiency in MANETs, D-ODMRP uses existing multicast destination node as forwarding node. In this protocol, the path from multicast source to multicast destination tends to use those paths passing through another multicast destination. In Figure 5, we have represented an example for one source and two receivers. In ODMRP protocol, paths P1 and P2 are selected, by default, for receivers R1 and R2, respectively. In D-ODMRP, R2 is nearer to R1 as compared to other receiver, so R2 can pick a route P2' via R1, as intermediate node and does not require any separate route.

A least additional cost is adaptable in case multiple paths are available. It also takes deferring time to calculate delay for reaching packet to the destination. It has simulated D-ODMRP for packet delivery ratio and control overhead and compared the results with that of ODMRP protocol over NS2 Simulator.

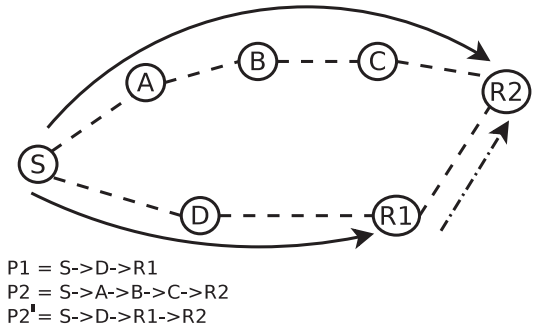

Fig. 5. D-ODMRP protocol.

3.2.13 LODMRP: Level-based On-Demand Multicasting Routing Protocol for MANETs [53]. Protocol tries to confine flooding of control packets within network by broadcasting only a part of these packets based on level-based approach. Each node decides to broadcast a Join Query packet based on its distance from the sender. The threshold for discarding join query is number of hops. Level represents the number of hops from sender to the node. Neighbor nodes transmit more packets as compared to faraway nodes. Control overhead, efficiency, and delay are analyzed with increasing number of sender and traffic load and compared results with ODMRP protocol over Glomosim Simulator.

3.3 QoS Metrics-based Extensions

Quality of Service parameters are not used to discover paths from source to destination, but to gratify the QoS requirements often given in terms of delay, congestion, bandwidth, and power. In this section, we have discussed briefly the improvements that have been made in ODMRP protocol proposed by different authors to ensure QoS support and reliable in case of route or link breakage. In Figure 4, we have listed QoS-based modification protocols on ODMRP.

In Figure 6, we have represented QoS metrics that have been used by researchers for enhancement in ODMRP protocol, to make it more robust, reliable, and reduce control overhead. QoS metrics contains following parameters such as SINR, Delay, energy, bandwidth, congestion, and so on. These metrics have been used to make efficient route and less prone to link failure due to high stability links. Researchers always try to refine QoS metrics to improve delivery ratio without degrading network throughput. A QoS metric can be used to enhance the quality of network according to application requirement, i.e., SINR metric is used to calculate the signal strength of a particular link that helps network administrator to increase the network performance.

3.3.1 QoS-LR-ODMRP: Quality of Service and Local Recovery for ODMRP Multicast Routing in Ad hoc Networks [69]. It proposes a new technique for supporting QoS routing in ODMRP by making acceptable estimation of available and required bandwidth with local route discovery. This protocol ensures that every node in the route setup should be based on bandwidth calculations for available bandwidth. It consumed node channel bandwidth that is provided by reserved bandwidth for flow on upstream and downstream neighbor of node. The protocol sets up route on the basis of available bandwidth of forwarding node. This protocol also proposes local route discovery on link breakage due to node mobility. It evaluates PDR and traffic admission ratio with increasing speed of node over Glomosim Simulator.

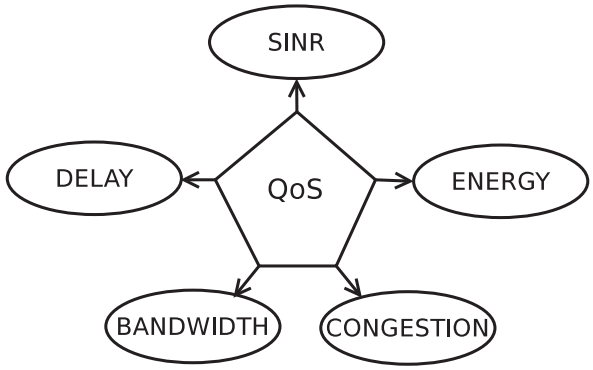

Fig. 6. QoS metrics.

3.3.2 A Cooperative Framework for Reliable Multicast Forwarding in MANETs [48]. It offers higher reliability and connectivity among multicast members in comparison to other existing reactive protocols. The author has proposed an innovative framework based on the cooperation between MAC and routing protocol. The proposed framework also adds some new features to ODMRP and IEEE 802.11 MAC layer for reliable forwarding by adding ODMRP with **D3MP (Dynamic Mesh-based Multicast MAC Protocol)** and **RRAR (Round Robin Acknowledge and Re-transmit)**. It evaluates signaling overhead, PDR, and Delay with variation in Multicast Group size on NS2 Simulator.

3.3.3 AAM-QoS: Agent-based Adaptive Multicast Routing with QoS Guarantee in MANETs [50]. This protocol guarantees QoS in terms of bandwidth, delay, jitter, and packet loss with agent-based adaptive algorithm. In this, a set of static and mobile agent moves around the network and collects the routing information. It involves clustering of nodes and selection of QoS-aware cluster head [72]. To satisfy multiple constraints, it identifies intermediate node and discovers multiple paths. It set up a QoS-aware path for the required multicast route. It evaluates packet delivery ratio and latency with mobility and group size on NS2 simulator.

3.3.4 IMRAANETs: An Improved Multicast Routing Algorithm in Ad hoc Network [49]. It analyzes the power variation of nodes to predict the topology change and link state. It calculates transmission power and rate of change of received power for any two intermediate nodes. It also calculates the response time to inform source about unreliable link/node to prevent route failure [84]. It reduces the route failure numbers and delivery delay without increasing extra overhead. This protocol compares response time with failure time to trigger the routing warning function. Simulations have been carried out for both ODMRP and Extended protocol to evaluate PDR with varying mobility speed.

3.3.5 TB-QoSODMRP: A Tree-based QoS Multicast Routing Protocol for MANETs [56]. It proposes a model that searches for QoS guaranteed path for a single source to set of destinations. In this, physical area is partitioned into equal-sized hexagonal cells, as shown in Figure 7, and a leader and backup leader are elected to maintain updated information about the topology. Position-based QoS MRP was proposed with GPS enabled on each node (device). The leaders are in the range of each node in the hexagonal cell. They find route on the basis of available bandwidth and delay to reach other intermediate nodes or destination. It is a type of group or cluster of nodes to transmit data effectively to each node by leader. It evaluates TBQMRP for CTRL packets transferred and

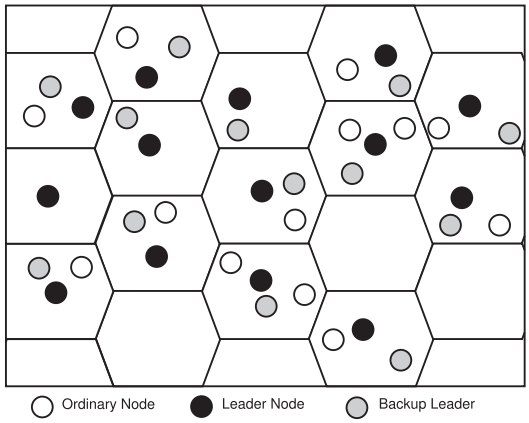

Fig. 7. Hexagonal cells in a scenario.

packet loss ratio with mobility speed on Glomosim Simulator. But, there is a drawback in case of leader and backup leader. All communications that are performed go under the leader node. So, the leader node has higher bandwidth and energy, else it would have gone through lack of resources.

3.3.6 ODMRP-IQoS: Providing Interference-aware Quality of Service Support for ODMRP [67]. Interference-aware QoS-ODMRP investigates bandwidth consumption under 2-hop interference model. It evaluates available bandwidth by employing a bit vector, named as Time Tag, to trace the transmission status within 2-hop neighbors and finds the clique (two or more nodes are in same transmission range) in network to avoid transmission interference. TTag is used for recording transmission status in most recent time from 1-hop neighbors. Nodes exchange TTAG with neighbors periodically for estimating bandwidth requirement. It analyzes delivery ratio and delay with increasing payload and shows improvement mainly with payload increment over Glomosim Simulator. Periodic transmission of TTag among neighbors is a major con for the protocol performance.

3.3.7 IODMRP: Improvement of Wireless Multicast Routing with Energy-efficiency Based on ODMRP [61]. It takes partial nodes in forwarding group that relay packets, and its choice is based on forwarder density and power state. Lesser the number of neighbor forwarding nodes, higher is the PDR. Some portion of forwarding group forwards the packet, where the portion is calculated on the basis of probability, $p(0 \leq p \leq 1)$. It calculates **power state (PS)** of node by dividing current received power by initial power and analyzes end-to-end delay and PDR of IODMRP over different number of receivers and maximum mobility speed over NS2 simulator.

3.3.8 DG-ODMRP: Delay-Guaranteed Multicast Routing in Multi-rate MANETs [63]. It estimates one hop delay and end-to-end delay based on varied transmission rates by monitoring the sensed busy-to-idle ratio of shared channel. Calculating both delays using IEEE 802.11 MAC is still challenging, because the radio channel is shared among neighbors. One-hop delay is sum of deferring and transmission time. Protocol senses busy to idle ratio of shared channel for 1-hop delay. The end-to-end delay can be determined by summing up all 1-hop delays in route. This approach also considers link with maximum signal rate. It compares 1-hop and end-to-end delay with DGMR and AQOR in single and multi-rate environment on NS2 simulator.

3.3.9 LSMRM: Link Stability Multicast Routing Protocol in MANETs [30]. Authors select stable forwarding node that is based on link connectivity with high stability in mesh-based MRPs. Stable route is selected by determining stable nodes that have high link quality in terms of estimated received power, bit error rate per packet, and distance between communicating nodes. They have maintained link stability database at every node. The drawback is that they have not given any mathematical and analytical model to prove or validate their implementations and results. They have improved PDR, delay, and routing overhead over changing multicast group size and transmission range and compared the obtained experimental results with ODMRP and EODMRP.

3.3.10 SRS-ODMRP: Stable Route Selection in ODMRP with Energy-based Strategy [58]. It forwards data on the basis of node energy. To select stable route, route expiration time and residual energy have been considered. Stable Weight-based method is used for ODMRP protocol to improve reliability. It calculates **Residue Energy (RES)** and **Route Expiration Time (RET)** and combines them to calculate shortest route. It appends position, direction, speed, and mobility. It analyzes end-to-end delay and control overhead with variation in mobility speed and multicast group size on OPNET simulator.

3.3.11 IQoS-ODMRP: A Novel Routing Protocol Considering QoS Parameter in MANETs [57]. It extends the ODMRP protocol to make it more suitable in disaster area network with group communication, adds QoS parameters such as bandwidth and delay in ODMRP, and takes consideration of mobility and analyzes the effect of time interval of sending packet with change in mobility. This protocol able to improves PDR and delay and compares the result with QoS-ODMRP on GlomoSim Simulator.

3.3.12 LLMR: A Link Stability-based Multicast Routing Protocol for Wireless Mobile Ad hoc Network [29]. This protocol finds the longer route (stable multicast route) in high mobility scenario. Authors have used weighted multicast routing algorithm to generate stochastic Steiner tree within **expected duration time (EDT)**, then applied learning automata-based approach to solve the problem. They have done extensive simulation on NS2 and compared the result with LSMRM and EODMRP protocols to validate the results. They have calculated route lifetime and PDR with changing host speed and multicast group size. It improves PDR, delay, and compares the result with QoS-ODMRP over Glomosim Simulator.

3.3.13 FA-QoSMRP: Fuzzy Agent-based QoS Multicast Routing in MANETs [40]. It provides the desired Quality of Service for user in group communication. A set of agents is used to operate in the following sequence: Creation of QoS Multicast mesh networks by using fuzzy inference system. A path to transmit the packet to receiver is selected from QoS Mesh. Mobile agents are employed to maintain QoS path. It analyzes PDR and control overhead and results reveal that FA-QoSMRP operates better than ODMRP on NS2 simulator.

3.3.14 MMRNS: Neighbor Support Reliable Multipath Multicast Routing in MANETs [28]. Authors have proposed a scheme for multipath multicast routing in MANETs using reliable neighbor selection. In this, a mesh is created from source to multicast destinations using maximum reliable pair factor of neighbors. In this algorithm, reliable pair factor depends upon energy and signal strength of node. Neighbor nodes are pruned by minimum threshold value and maintain route against node/link failure. Authors have analyzed their results with ODMRP and EODMRP protocols in terms of PDR and control/computation overhead in respect of number of nodes and groups with mobility considerations. Figure 8 shows random network topology consisting of one source(S) and two destinations(R1 and R2). In this figure, MMRNS picks up two reliable paths for each destination at any time t. First, the source transmits data through higher priority level path P1 to

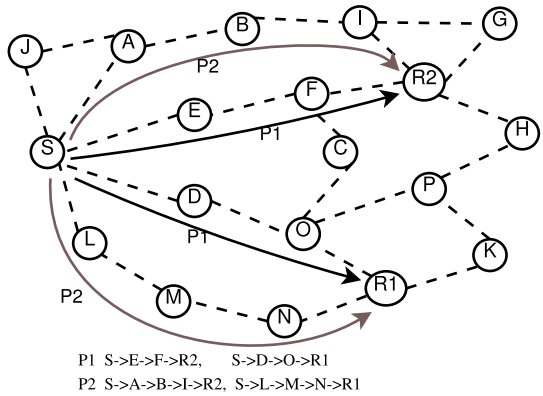

Fig. 8. Multiple multicast using reliable network selection.

destinations R1 and R2. In multicasting, load of the transmission has been increased on single node/link due to multiple transmissions. So, using different priority paths, load can be reduced by transmitting data through multiple paths like P2.

3.3.15 On-Demand Multicast Routing Protocol with Efficient Route Discovery [59]. In this, limited flooding in ODMRP reduces the packet overhead drastically by sending JQ messages from only delay satisfaction nodes. It calculates 1-hop delay for every node by summing up transmission delay, contention delay, and queuing delay. Node selects minimum hop delay node to transmit the data and only floods Join Query messages. It analyzes delay, overhead, and PDR with increasing number of multicast receivers over NS2 simulator and compares results with ODMRP and EODMRP protocol.

3.3.16 QoS-MRPM: QoS-based Multicast Routing Protocol in MANETs [44]. QoS-based MRP provides stable multicast paths with enough bandwidth. In this, entropy is treated as an important parameter to find stable path. Protocol uses bandwidth reservation mechanism to achieve QoS. It can be used to select stable path with longer lifetime [65]. It compares the results of Average Delay and PDR with varying Velocity of Sending Packet for ODMRP over NS2 simulator.

3.3.17 Extending ODMRP for On-Site Deployments in Disaster Area Scenarios [45]. This work extended the ODMRP protocol to make it more suitable in disaster area network with group communication. It is link quality-based routing protocol that requires Hello packet transmissions. First, they prioritize control messages and use Overhead Reduction Mechanism to provide better throughput in disaster areas. It evaluates packet loss ratio for GPS packet per node over NS2 simulator and tested it on Physical testbed.

3.3.18 NPA-MAM: New Power-aware Multicast Algorithm for Mobile Ad hoc Networks [60]. Power-aware multicast routing algorithm uses the residual battery life for multicasting from source to a group of destinations. The proposed protocol considers residual energy as a QoS metric while forwarding the data packets. The proposed model chooses a node with maximum remaining power among all the nodes. It extends the network lifetimes of the node and the network without degrading the network throughput. It compares the results with network lifetime, control bytes per data, and PDR with varying group size on NS2 simulator.

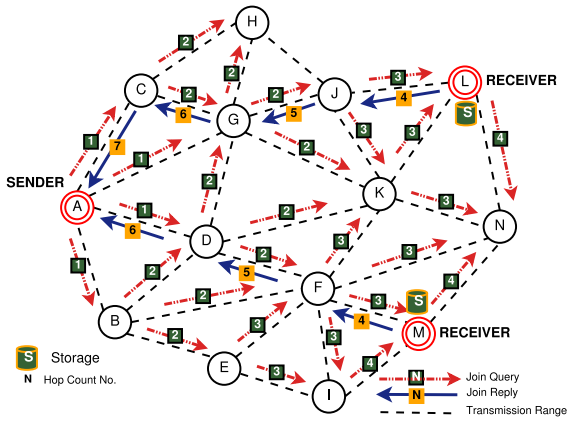

Fig. 9. LSMRP workflow.

3.3.19 LSMRP: Link Stability Multicast Routing Protocol in MANETs [1]. SINR metric is used for link stability estimation over ODMRP protocol. This algorithm determines reliable path to reduce link failures and re-routing overhead. It simply estimates a link that sustains for longer duration in the network. Link failure may occur due to mobility of nodes. Higher signal strength links have higher probability of existence. It increases route lifetimes without degrading the network throughput. Here, authors have performed extensive simulations with increasing mobility and multicast group size on Exata simulator and improved performance in terms of PDR and average End-to-End Delay.

Link stability of each individual link throughout the route has been estimated. To estimate the best suitable link or stable link, requesting route packets have been buffered for a threshold time. Otherwise, ODMRP and other similar protocols are only entertaining the FCFS request. This approach chooses a link from multiple options based on RSSI indicator, as it is more sustainable for required time period. Route request and route reply messages have been used for route establishment and presented in the Figure 9.

ODMRP protocol uses hop count to select the routes for multicast group address. LSMRP has been using RSSI indicator for stable route selection. This work has been extended for multiple QoS metrics to estimate the reliable links and routes for single source to multiple destinations [16].

Table 1 represents the summary of mesh topology-based MRPs in Mobile ad hoc networks. In this table, protocols are classified in the following categories: modification components, features, techniques, and improvement parameters. In this table, "–" represents absent of the feature or function and "✓" represents presence. In simulator, the numbers 1, 2, 3, 4, and 5 represent simulator names NS2, Qualnet, Glomosim, OPNET, and EXata/Cyber 2.0, respectively.

4 INTRODUCTION TO LINK STABILITY IN MULTICAST ROUTING PROTOCOL

Quality of Service is calculated on the basis of link stability [29]. Fluctuating link stability in wireless networks has a fundamental impact on network performance [81]. Link stability is subject to less robust link/node failure, node mobility, and low reliability. Researchers try to make efficient routing protocols that are able to deal with link unreliability. Route persists for longer duration because stable links are en-routed. As a result, reduced computations due to less re-routings lead to reduced overhead. Here, we give a brief overview of requirement of link stability and its metrics.

Table 1. Summary of Mesh-based Multicast Routing Protocol

Protocol Name	ODMRPMPR [21]	R-ODMRP [43]	EODMRP [27]	AMRPWMN [51]	LF-RDTODMRP [20]	ODMRP-LR [18]	RB-MRMAM [68]	ODMRP-RR [47]	ODMRP-DLJQ [54]	PRIME [66]	RI-ODMRP [55]	DODMRP [42]	LODMRP [53]	QoS-LR-ODMRP [69]	ODMRP-IQoS [67]	DG-ODMRP [63]	AAM-QoS [50]	IMRAANETs [49]	IQoS-ODMRP [57]	SRS-ODMRP [58]	TB-QoSMRP [56]	MMRNS [28]	FA-QoSMRP [40]	NPA-MAM [60]	QoS-MRPM [44]	Moralism [6]	Route-Stability [17]	MCLSPM [16]
Year of Publication	2003	07	08	09	09	09	09	10	10	12	11	12	13	08	09	09	10	10	10	10	11	12	12	13	12	16	17	18
Modification Components:																												
Forwarding Group	✓	✓	✓	–	–	–	✓	✓	✓	–	–	–	✓	–	–	–	–	–	–	–	–	–	–	–	✓	–	–	✓
Routing Selection	✓			✓	–	–	–	–	✓		✓	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Features:																												
Local Route Repair	–	–	✓	–	–	✓	–	–	–	–	–	–	–	✓	–	–	–	–	–	–	–	–	–	–	–	–	✓	–
Receiver Joining	–	–	✓	–	–	✓	–	–	–	✓	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	✓	–	–
Dynamic Timer Adaption	–	–	✓	–	–	–	–	–	–	–	–	–	✓	–	–	–	–	–	✓	–	–	–	–	–	–	✓	–	–
Mobility Handling	–	–	–	–	✓	–	–	–	–	✓	–	–	–	–	–	–	–	–	✓	✓	–	✓	–	–	–	✓	–	✓
Link Asymmetry	✓	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	✓	✓	✓
Energy	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	✓		✓	✓	✓	✓	✓	–	–	–	–
Delay	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	✓	✓		✓	✓	–	–	✓	–	–	–	–	–
Signal Strength	–	–	–	–	–	–	✓	–	–	–	–	–	–	–	–	✓	–	–	–	–	–	✓	–	–	–	✓	–	✓
Bandwidth Reservation	–	–	–	–	–	–	✓	–	–	–	–	–	–	✓	✓	–	✓	–	✓	–	–	–	–	–	–	–	–	✓
Congestion/Interference	–	–	–	–	–	–	–	–	–	–	–	–	–	✓	✓	–	–	–	–	–	–	–	–	–	–	–	–	✓
Application feedback	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	✓	–	–	–	–	–	–	–	–	–
Techniques:																												
Periodic Hello	✓	–	–	✓	–	✓	–	–	–	–	–	–	–	✓	✓	–	–	–	✓	–	–	–	–	–	–	–	✓	–
Link Quality	–	–	✓	–	–	✓	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	✓	–	–	–	✓	–	✓
Passive Ack	✓	–	✓	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	✓	✓	✓
Join Query Delaying	–	–	–	✓	–	–	–	–	–	–	–	✓	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	✓
Route Discovery Suppression	–	–	–	✓	✓	–	–	–	✓	–	–	–	✓	–	–	–	–		–	–	–	–	–	–	✓	✓	–	–
Conserving FG Joining	–	–	–	–	✓	–	–	✓	–	–	–	✓	–	–	–	–	–	–	–	–	–	–	–	–	–	✓	–	–
Improvement Parameter:																												
PDR	✓	✓	✓	✓	✓	–	✓	✓	–	✓	✓	–	✓	✓	✓	–	✓	✓	✓	–	–	✓	✓	–	✓	✓	✓	✓
Delay	–	–	–	–	✓	–	✓	–	✓	✓	–	–	✓	–	✓	✓	✓	–	✓	✓	–	–	–	✓	✓	✓	✓	✓
Overhead	–	–	–	–	✓	✓	–	–	✓	–	–	✓	✓	–	–	–	–	–	–	✓	✓	✓	–	–	–	–	–	✓
Category	A	A	A/B	A/B	A/B	B	C	A	B	A	A	B	B	A/C	C	C	C	C	C	C	C	C	C	C	C	C	C	C
Simulator	1	1	3	2	2	2	1	1	2	3	1	1	2	2	2	1	1	1	2	4	2	1	1	1	1	5	5	5

(✓ Supported and – Not_Supported) (A=Routing-based, B= Attack-based and C= QoS-based).

4.1 Requirement of Link Stability

In this section, we are going to discuss the requirement or benefits of adding the concept of link stability in the route construction [73]. We have listed some of them below:

- Energy Efficient. Energy can be conserved by reducing number of reformations, i.e., selecting a stable route. Node energy is wasted due to broadcast of request and reply packets for repeated demand of route in case of multiple links failure.
- Accuracy. It is essential to estimate a longer/stable path from single discovery to ensure that link would not break for any reason.
- Reactivity. Due to high mobility in the network, our protocol should be well adaptive to every small change in the network.
- Stability. Stable route in terms of prolonged persistence should be selected. Nodes at minimum distance and with maximum residual power are regarded as more stable than others.

There is a flaw between stability and reactivity, as they can not be performed together, because both are opposite to each other. So, we have to make reactive protocols that satisfy both conditions: stabilization and reactivity.

4.2 Link Stability Metrics

Link stability metrics help us to find QoS-aware links. Metrics that affect the link stability are as follows:

- Signal Strength or SINR or BER,
- Transmission Delay or ETX,
- Residual Energy or Power,
- Bandwidth Reservation,
- Congestion or Interference.

4.2.1 Signal Strength/SINR/BER. SINR is the power of certain signal of interest divided by sum of interference power from other signal and background noise. SINR is estimated from signal strengths between nodes for transmitting data.

$$\text{SINR} = \frac{Rec_signal_Power}{other_Interference + background_noise} \tag{1}$$

Bit Error Rate (BER) can be decided from number of bits dropped out of total number of bits sent to the destination node. SINR is used to find stable link that has more lifetime than other nodes. We can also take continuous SINR values to predict the direction or mobility of node.

4.2.2 Delay/ETX. Total Delay can be calculated by summing up the Transmission delay (packet transfer time between nodes), Queuing delay (packet has to wait in queue for getting sequenced), and Processing delay (after queuing time for transferring the packet or wait for channel).

$$Delay = TD + QD + CD, \tag{2}$$

where TD is Transmission_Delay, QD is Queuing_Delay, and CD is Contention_Delays. Transmission delay is the time taken to transmit a packet from one hop to next hop. In QoS metric, link with lower transmission delay will be included in the route for establishing stable path. Queuing delay and processing delay are approximately equal to all, or otherwise it depends upon the type of application being executed.

4.2.3 Residual Energy/Power. Energy has always been a major concern in MANETs. We are calculating node residual energy and energy consumption for n transmissions. Most of the time, we assume that all nodes have same transmission power. Node that has more residual energy and less power consumption in data transmission is selected as intermediate node for transmission on route.

$$\text{Power Ratio} = \frac{Remaining\ Power}{Total\ Capacity} \tag{3}$$

Power ratio tells us about node's remaining power. We can calculate power ratio by dividing remaining power by total capacity and conclude that it lies between two ranges: Low range and High range. Values above threshold come in High power range.

4.2.4 Bandwidth Reservation. Network bandwidth reservation is used to identify the capacity of participating node and the corresponding link. We require minimum bandwidth to communicate data for QoS-aware routing protocols. Available bandwidth of a node can be estimated through the idle time of channel and transmission range. We can calculate idle time of channel by monitoring the node, whether it is busy or not in receiving or sending any packet. Bandwidth reservation is compulsory in multimedia applications for improving delay and jitter in data streaming. For better analysis, we have to precisely estimate the network bandwidth and total bandwidth consumption in transmission from requesting applications.

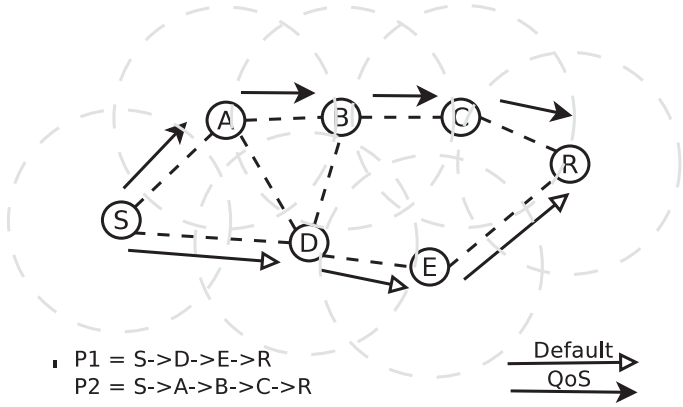

Fig. 10. QoS_route construction.

4.2.5 Congestion. Congestion can be estimated in terms of interference and load on a node. We can calculate path encounter for every node to estimate congestion on a link or a node. Path encounter can be detected when a node comes in the transmission range of participating node. Path encounter value is the number of nodes in the transmission range [9] or number of nodes that are affecting (consuming) network bandwidth of participating node. We can select node with minimum path encounter value at the time of route selection. Occurrence of congestion depends on various parameters. It can be related to bandwidth consumption and conflict due to simultaneous transmissions.

In Figure 10, we have represented a QoS-aware route and a default route construction. We have two routes P1 and P2 as Default and QoS-aware routes, respectively. We can use QoS metrics to calculate stable link for transmitting data. In basic ODMRP protocol, receiver R can be reached via nodes D and E as forwarding group member, due to less hop count and fastest response from P1 route. But, we can not say that it is a stable route or not. We can use QoS metric to calculate link availability time for finding stable and robust link for route construction, as shown in P2. Although, route P2 has increased number of hops, entire route would remain stable till route expiration time. We can also used route stability in place of link stability for the reliable communication between source and destination. A route stability is a combination of stable links to ensure the uninterrupted communication between two end devices, and application requirements help us to select the suitable metric.

5 DESIGN ISSUES AND CHALLENGES IN MULTICAST ROUTING PROTOCOL

Although researchers have designed numerous MRPs and techniques, there remains a number of open issues and challenges. All available protocols provide mechanism to improve the performance of ODMRP protocol. However, each of these approaches suffer from certain issues that decide the preciseness of the approach. Some of the issues [71] are listed as follows:

(i) ***Energy Efficiency:*** Mobile nodes typically run on limited energy resources, so it is required to design energy preserving protocols for group communication. Low throughput and high interferences over wireless channels is due to high energy consumption in MANETs.

We can reduce energy consumption by decreasing number of nodes that are included in forwarding group and selecting the nodes that have highest energy to transmit the datagram to make equal consumption by every node. We can also preserve energy through broadcast communication.

Many protocols have been proposed for energy conservation [61]. If we select a high-energy link in advance, then energy would not be consumed due to retransmission of packet at MAC layer. Energy issues increase with high mobility, high contention because we are using shared channel-based MAC protocol.

(ii) ***Robust and Reliable Multicasting:*** Due to arbitrary movement of mobile nodes, link failures are usual in MANETs. MRP should be resistive to mobility and gain high PDR. Reliable multicasting ensures that data from source node should reach every destination with ditto set of messages.

In MANETs, the reliability of multicast frames cannot be guaranteed, because it depends upon mobility, multicast group size, and traffic load. In MRPs, there are no RTS/CTS control frames to inquire about the availability of channel. Moreover, there is no provision of acknowledgement to achieve reliable communication. Unreliability in the network increases due to transmission of real-time multimedia traffic. A node is unstable or unreliable due to its high and unpredictable mobility.

(iii) ***Efficiency and Control Overhead:*** Efficiency can be defined as ratio of total number of packets received to total number of packets transmitted at receiver. Total number of control packets transmitted in the entire network to maintain routes to multicast group signify control overhead. Bandwidth consumption at a node is higher due to control packets such as transmission of Hello packets, route request packets, and so on.

(iv) ***QoS-aware Multicasting:*** QoS is achieved by set of service parameters during data streaming over multicast group from a source. QoS attributes such as delay, bandwidth, probability of packet loss, signal strength, and so on, are vital to get enhanced performance in terms of PDR and end-to-end delay. It facilitates reduced number of route reconfigurations in the network in case of link or route failure.

A large number of approaches with QoS support have been published for mesh-based protocol. Due to highly dynamic topology of network, providing QoS support is very difficult. A QoS modification can be executed at different layers according to the application requirements. At MAC layer, we can determine delay and packet loss ratio. Similarly, we can obtain received signal strength, bit error rate, and transmission quality at physical layer.

Real-time video streaming requires a minimum bandwidth to communicate, so we can add QoS parameter to enhance transmission quality and bandwidth assurance.

(v) ***Secure Multicasting:*** In MANETs, secure networking has become a subject of great concern to researchers, because wireless networks are more prone to passive and active attacks. In multicast scenario, security is more sophisticated due to number of receivers attached to the network. A single attacker node can degrade the performance of entire network.

The MRPs should be efficient to provide protection from denial of service attacks, misbehaving nodes, unauthorized access to data, and so on. We can make MANETs secure from unauthorized access of data by applying encryption mechanisms with group key management; however, we have to mitigate excessive overhead that would be generated due to cryptography techniques. In addition, mobile nodes run on limited energy resources and have low computing power, thus, applying such complex techniques would drain off available resources.

In MANETs, security is receiving additional attention due to infrastructure-less network, no central administration, dynamic topology, and so on. Several solutions have been proposed for security in MANETs, but not many lightweight mechanisms. Mainly, approaches are based on delay calculation, behavior analysis, trust-based, and geo-casting. Geo-casting is used for calculating node position in battlefield to check authenticity of node, because single node can be caught and made malicious. Thus, we require a flexible and high security

mechanism that can adapt in all conditions discussed above. All these conditions are difficult to implement to secure MRPs for multicast communication.

6 PROPOSED SOLUTIONS TO OVERCOME THE ABOVE CHALLENGES

To overcome all the challenges and issues in the existing MRPs, three protocols have been proposed for reliable communication in temporary networks such as wireless Ad hoc network, Internet of things [80], and Mobile Ad hoc networks for multimedia data. In the first work (Moralism [6]), a device mobility model has been presented for predicting the node position in the coming future time in the Ad hoc networks. We are predicting the suitable node with maximum active time during the next communication and selected for next communication. Changes in RSSI value will help to determine the node position whether it is coming or going by estimating node's communication range. Signal strength is considered as a QoS metric to calculate link stability for route construction.

In the second proposed work (Route Stability [17]), both link and route stability parameters have been estimated for constructing a reliable route for particular communication, because last approach will not be feasible in case of route stability. A combination of reliable links cannot ensure a reliable route in Ad hoc networks. Number of successful receiving of periodic packets at expected nodes will be able to calculate the link stability. Route stability is the combination of different stable link for the desired path. After applying route stability parameter for node selection, significant improvement in throughput is achieved. The link active duration is directly proportional to the periodic packets receiving probability.

After finding the limitation of previous proposed work, a group of parameter (constraints) has been used for estimating a reliable link for the communication in Reference [16]. This third protocol (MCLSPM) has contention count, signal strength, and hop count for finding the suitable link for the forwarding data. Finally, **LSF (Link stability factor)** function will combine all these normalized parameters into a single function. This approach will find a route that is not too dense or not to sparse with minimal number of hop count. This approach will help to overcome all the disadvantages of the above proposed works.

6.1 Results

We have described the proposed protocols for smooth and reliable communication in between mobile nodes. In this subsection, all protocols are compared with the existing works for data transmission in a multicast networks. The comparisons (PDR, AEED) between the proposed routing protocols over generalize simulation parameters have been presented.

PDR is the number of successfully transmitted packets to the destinations. Figure 11 shows the comparison between proposed protocols and existing protocols in terms of PDR with increasing the network mobility. This clearly depicts that proposed approaches (MCLSPM, Moralism, and Route Stability) are providing better throughput as differentiate to old routing protocols (ODMRP [26], LSMRP [1], SMR [73], MMRNS [28]) at every node mobility speed. Increase in mobility speed will decrease the PDR due to increase in network instability. The Moralism work is not consistent for high movement network, because there is significant drop in PDR after 20 m/s, as presented in the graph. **Multi-constraint objective (MCLSPM)**, which helps to estimate the stable route for active route time, supplies the highest number of packets reception as compared to other protocols.

The **average end-to-end delay (AEED)** indirectly shows the network response time for transmission of data and control packets. A reliable protocol should always provide less jitter and high throughput. Comparison between routing protocols has been shown in the Figure 12. Higher AEED will slow down the network and increase the packet drop ratio due to buffer overflow. In most

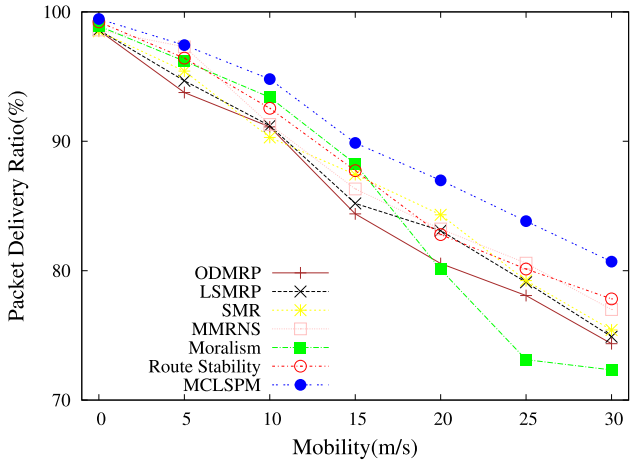

Fig. 11. Packet delivery ratio (%) vs. node mobility.

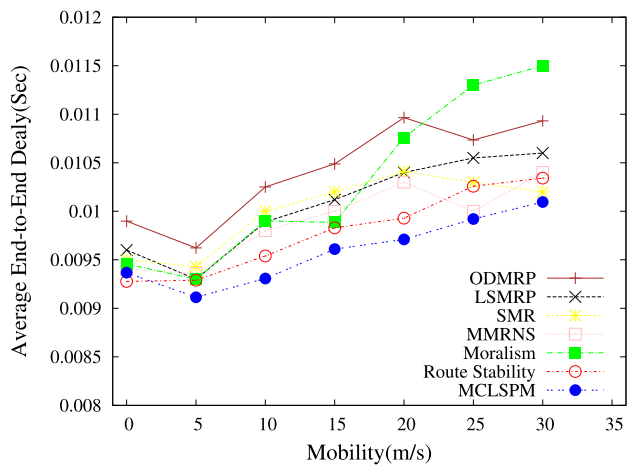

Fig. 12. Average end-to-end Delay (sec) vs. node mobility.

scenarios, MCLSPM provides better AEED as compared to other existing proposal protocols due to stable route for active route duration. This will lead to cost reduction in terms of expensive route reconfiguration before the next route request.

We have also tested our proposed protocols on **Flying Ad hoc networks (FANETs)** [74]. Here, the scenarios have additional challenges of third dimension, speed, distance. Proposed approach (MCLSPM) has outperformed other routing protocols in terms of each performance metric. For experiment, a simulated network scenario has been created on 3D map of EXata Cyber [41] network emulator by considering FANETs parameters.

7 FUTURE DIRECTIONS FOR MULTICAST ROUTING PROTOCOL IN MANETS

We believe that investigations for employing multicast communications in MANETs is still in its developing stage. It is mandatory to improve the protocol to be more robust and reliable in case

of link failure or dynamic changes in topology of the network. We have defined open issues, such as QoS guarantee, reliability, power efficiency, and security provisioning, that are essential for an effective routing protocol. It is difficult to design an MRP that successfully takes all these issues into consideration. We can also refine link stability parameters, which are defined in the last section, to improve the protocol performance. We can combine different routing modification mechanisms to make more resilient protocols.

On the basis of our understanding of multimedia data transfer over multicast MANETs, following research challenges may be taken up as an extension in the future research.

(1) Researchers can work to consolidate a system using our approaches to establish uninterrupted communication in a real-time application.
(2) Develop a robust QoS-aware MRPs that can reliably transmit video streaming data over flying Ad hoc networks [7]. Our protocol should be reliable with all the constraints in FANETs/Internet of Drones [77] network, such as scalability and high bandwidth.
(3) Securing the multimedia data communication between multicast nodes from the intruder for any kind of data breaches in mobile networks.
(4) Reliable QoS-aware techniques can be directly transferred in other similar networks, such as Internet of things, Software defined networks, 5G, and IoD.
(5) Researchers can use AI and ML approaches to make it more scalable and secure.

8 CONCLUSION

This article discusses the state-of-the-art research in mesh-based MRPs in edge-based ad hoc networks. From discussions as presented earlier, it can be inferred that selecting QoS metric for the specific problem domain is significant especially in MRP. A suitable QoS metric is useful in assessing "goodness" of a routing solution as per requisite performance. Various enhancements in ODMRP have been discussed on the basis of routing modifications and Quality of Service parameters. The article categorizes the protocols on the basis of type of modifications to achieve better throughput in terms of packet delivery ratio, end-to-end delay, control overhead, and packet loss ratio. A critical review of existing MRPs has been presented, and each protocol is discussed with its advantages and limitations. Also, the issues regarding multicast routing protocols in MANETs has been discussed. This study will give a complete background to the researchers who want to pursue their work in this domain.

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